



Aishu R 251001007-2025-IT ▾

**A2****Started on** Monday, 24 August 2026, 12:31 PM**State** Finished**Completed on** Monday, 24 August 2026, 12:32 PM**Time taken** 1 min 1 sec**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:

The first line contains the no of elements in the list-n

The next n lines contain the elements.

Output:

Sorted list of elements

**For example:**

| Input          | Result         |
|----------------|----------------|
| 5              | 12 34 67 78 98 |
| 67 34 12 98 78 |                |

**Answer:**

```

1  #include <stdio.h>
2
3  void quickSort(int a[], int low, int high)
4  {
5      int i, j, pivot, temp;
6
7      if (low >= high)
8          return;
9
10     pivot = a[low];
11     i = low + 1;
12     j = high;
13
14     while (i <= j)
15     {
16         while (i <= high && a[i] <= pivot)
17             i++;
18
19         while (a[j] > pivot)
20             j--;
21
22         if (i < j)
23         {
24             temp = a[i];
25             a[i] = a[j];
26             a[j] = temp;
27         }
28     }
29
30     temp = a[low];
31     a[low] = a[j];
32     a[j] = temp;
33
34     quickSort(a, low, j - 1);
35     quickSort(a, j + 1, high);
36 }
37
38 int main()
39 {
40     int n, i;
41     int a[1000];
42
43     scanf("%d", &n);
44
45     for (i = 0; i < n; i++)
46         scanf("%d", &a[i]);
47
48     quickSort(a, 0, n - 1);
49
50     for (i = 0; i < n; i++)
51         printf("%d ", a[i]);
52     printf("\n");
53 }
```

```
47 |  
48 | quickSort(a, 0, n - 1);  
49 |  
50 | for (i = 0; i < n; i++)  
51 | {  
52 |     if (i > 0)
```

|   | Input                               | Expected                      | Got                           |   |
|---|-------------------------------------|-------------------------------|-------------------------------|---|
| ✓ | 5<br>67 34 12 98 78                 | 12 34 67 78 98                | 12 34 67 78 98                | ✓ |
| ✓ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✓ |
| ✓ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90    | 1 2 3 4 5 6 7 8 9 10 11 90    | 1 2 3 4 5 6 7 8 9 10 11 90    | ✓ |

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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